

Problem 1 (p.200#1). Prove the formula of Gauss:

$$(2\pi)^{\frac{n-1}{2}} \Gamma(z) = n^{z-\frac{1}{2}} \Gamma\left(\frac{z}{n}\right) \Gamma\left(\frac{z+1}{n}\right) \cdots \Gamma\left(\frac{z+n-1}{n}\right).$$

Solution. Observe that

$$\begin{aligned} \sum_{i=0}^{n-1} \frac{d}{dz} \left(\frac{\Gamma'(\frac{z}{n} + \frac{i}{n})}{\Gamma(\frac{z}{n} + \frac{i}{n})} \right) &= \sum_{i=1}^n \sum_{k=1}^{\infty} \frac{1}{(\frac{z}{n} + \frac{i}{n} + k)^2} \\ &= n^2 \sum_{i=1}^n \sum_{k=1}^{\infty} \frac{1}{(z + nk + i)^2} = n^2 \sum_{m=1}^{\infty} \frac{1}{(z + m)^2} = n^2 \frac{d}{dz} \left(\frac{\Gamma'(z)}{\Gamma(z)} \right) \end{aligned}$$

so

$$\Gamma\left(\frac{z}{n}\right) \Gamma\left(\frac{z+1}{n}\right) \cdots \Gamma\left(\frac{z+n-1}{n}\right) = e^{az+b} \Gamma(z)$$

for some constants a and b . Taking $z = 1$ and $z = 2$, we get

$$\begin{aligned} \Gamma\left(\frac{1}{n}\right) \Gamma\left(\frac{2}{n}\right) \cdots \Gamma\left(\frac{n-1}{n}\right) &= e^{a+b} \\ \frac{1}{n} \cdot \Gamma\left(\frac{1}{n}\right) \Gamma\left(\frac{2}{n}\right) \cdots \Gamma\left(\frac{n-1}{n}\right) &= e^{2a+b} \end{aligned}$$

so $e^a = \frac{1}{n}$ and

$$e^b = n \cdot \Gamma\left(\frac{1}{n}\right) \Gamma\left(\frac{2}{n}\right) \cdots \Gamma\left(\frac{n-1}{n}\right) = n \sqrt{\prod_{k=1}^{n-1} \Gamma\left(\frac{k}{n}\right) \Gamma\left(1 - \frac{k}{n}\right)} = \frac{n\pi^{\frac{n-1}{2}}}{\sqrt{\prod_{k=1}^{n-1} \sin(\frac{\pi k}{n})}}.$$

Writing $\sin(x) = \frac{1}{2i}(e^{ix} - e^{-ix})$, we get

$$\prod_{k=1}^{n-1} \sin\left(\frac{\pi k}{n}\right) = \frac{1}{(2i)^{n-1}} \prod_{k=1}^{n-1} (e^{\pi ki/n} - e^{-\pi ki/n}).$$

Letting $\omega = e^{2\pi i/n}$, we have

$$(e^{\pi ki/n} - e^{-\pi ki/n})(e^{\pi(n-k)i/n} - e^{-\pi(n-k)i/n}) = -2 \cos(\pi(n-2k)/n) - 2 = \omega^k + \omega^{-k} - 2.$$

This means that the square of the product of the $e^{\pi ki/n} - e^{-\pi ki/n}$ is equal to

$$\prod_{k=1}^{n-1} (\omega^k + \omega^{-k} - 2) = \prod_{k=1}^{n-1} \frac{(\omega^k - 1)^2}{\omega^k} = \frac{n^2}{\omega^{n(n-1)/2}}.$$

Knowing that the product of the sines is a positive real number, we can conclude that it is equal to $\frac{n}{2^{n-1}}$, so

$$e^b = n^{\frac{1}{2}} (2\pi)^{\frac{n-1}{2}},$$

which proves the desired identity after substituting in the values of e^a and e^b and rearranging. $\square$

Problem 2 (p.200#2). Show that

$$\Gamma\left(\frac{1}{6}\right) = 2^{-\frac{1}{3}} \left(\frac{3}{\pi}\right)^{\frac{1}{2}} \Gamma\left(\frac{1}{3}\right)^2.$$

Solution. By Legendre's duplication formula applied to $z = \frac{1}{6}$ and the identity for $\Gamma(z)\Gamma(1-z)$,

$$\sqrt{\pi}\Gamma\left(\frac{1}{3}\right) = 2^{-\frac{2}{3}}\Gamma\left(\frac{1}{6}\right)\Gamma\left(\frac{2}{3}\right) = 2^{-\frac{2}{3}}\Gamma\left(\frac{1}{6}\right)\frac{2\pi}{\sqrt{3}\cdot\Gamma(\frac{1}{3})},$$

which gives us what we want after rearranging. □

Problem 3 (p.200#3). What are the residues of $\Gamma(z)$ at the poles $z = -n$?

Solution. Looking at the product expansion of $\Gamma(z)$ shows that its poles are all simple. For an arbitrary nonnegative integer n ,

$$\lim_{z \rightarrow -n} (z+n)\Gamma(z) = \lim_{z \rightarrow -n} (z+n) \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n)} = \lim_{z \rightarrow -n} \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n-1)} = \boxed{\frac{(-1)^n}{n!}}. \quad \square$$

Problem 4. Show the convergence of the series of γ , namely

$$\gamma - 1 = \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{1}{n+1} - \log(N+1) \right).$$

Solution. Write the argument of the limit as

$$\begin{aligned} \sum_{n=1}^N \frac{1}{n+1} - \log\left(1 + \frac{1}{n}\right) &= \frac{1}{N+1} - \log 2 + \sum_{n=2}^N \frac{1}{n} - \log\left(1 + \frac{1}{n}\right) \\ &= \frac{1}{N+1} - \log 2 + \sum_{n=2}^N \left(\frac{1}{n} - \frac{1}{n} + \frac{2}{n^2} - \frac{3}{n^3} + \cdots \right) \\ &= \frac{1}{N+1} - \log 2 + \sum_{n=2}^N \left(\frac{2}{n^2} - \frac{3}{n^3} + \frac{4}{n^4} - \cdots \right). \end{aligned}$$

The n th summand in the sum at the end is nonnegative and bounded above by $\frac{2}{n^2}$, so its limit as N goes to infinity converges by comparison with $\sum_{n=2}^{\infty} \frac{2}{n^2} = \frac{\pi^2}{3} - 2$, hence

$$\lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{1}{n+1} - \log(N+1) \right) = \sum_{n=2}^{\infty} \left(\frac{2}{n^2} - \frac{3}{n^3} + \frac{4}{n^4} - \cdots \right) - \log 2$$

exists. □